



Recommending Usability Improvements with Multimodal Large Language Models

Jinhyeok Choi

2026.07.19

Base Lab @DGIST



Overview

1. Introduction

2. Approach

3. Study Design

4. Results

5. Discussion

6. Threats to Validity

Introduction

Usability → describes quality attributes of application user interfaces that determine how effectively users can interact with them

Good Usability



Help with the acceptance and usage of a system

vs

Poor Usability

lower user satisfaction, reduced adoption, and higher costs for support and maintenance



Introduction

Traditional Usability Evaluation

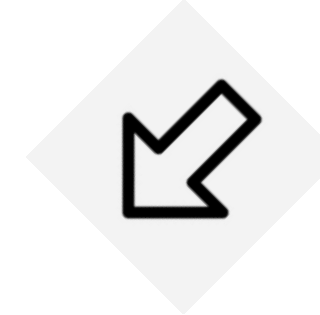
1. Usability Testing

Real users are observed while solving tasks



2. Usability Inspection

Experts simulate the task-solving of real users



Both aim to identify **usability issues** but are **resource-intensive**, as they require manual analysis, expert knowledge, and dedicated time

10 Usability Heuristics



Visibility

Show system status, tell what's happening



Freedom

Provide good defaults & undo



Error Prevention

Helps users avoid making mistakes



Flexibility

Make advanced tasks fluid and efficient



Error Recovery

Helps Users recognize, diagnose and recover from errors



Mapping

Use familiar metaphors & language



Consistency

Use same interface and language throughout



Recognition

Make information easy to discover



Minimalism

Provide only necessary information in an elegant way



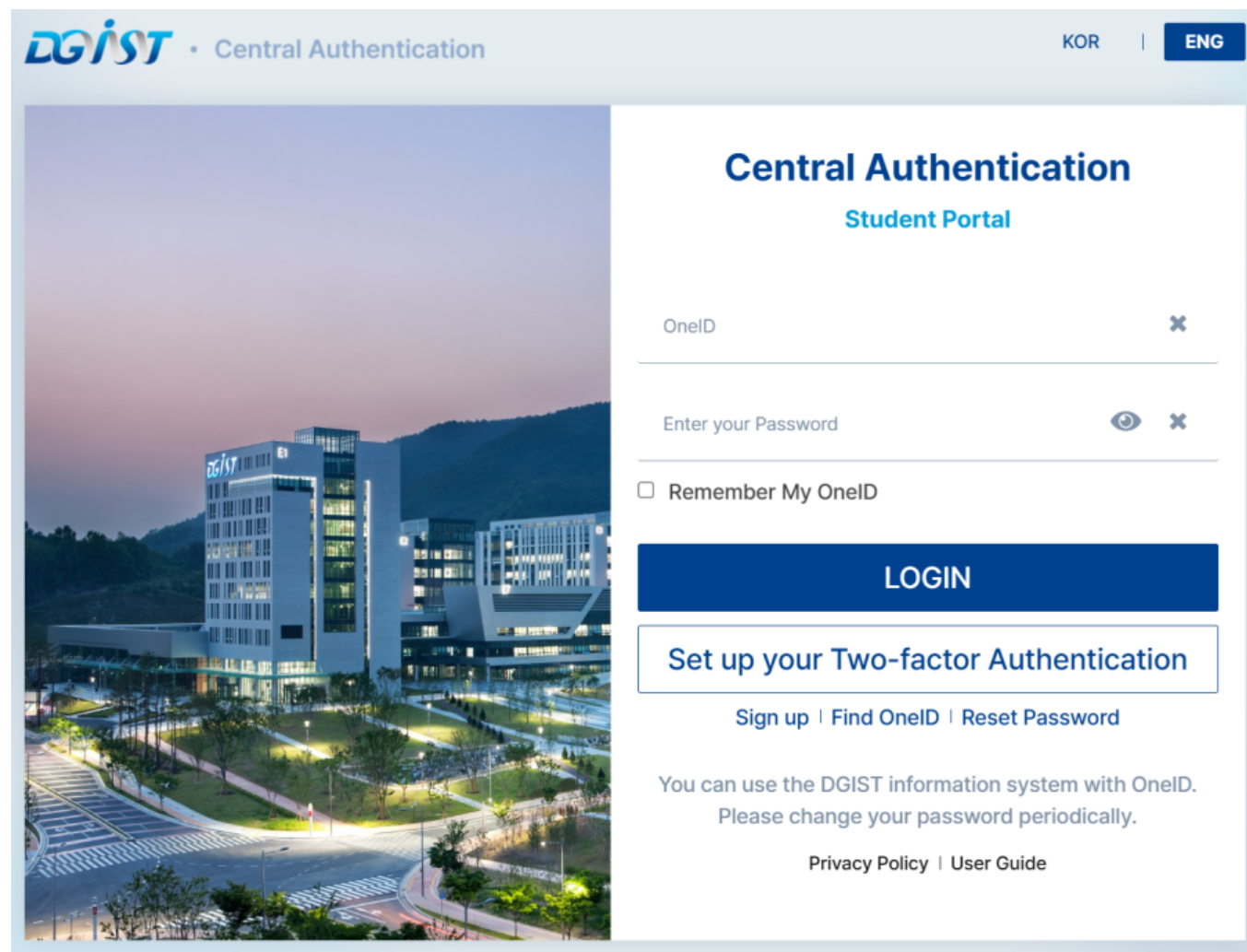
Help

Use proactive and in-place hints to guide users

Introduction

MLLM(Multimodal Large Language Model)

Visual Inputs



Textual Context

App Description: DGIST Login Page

Task: given ID, PW → Login



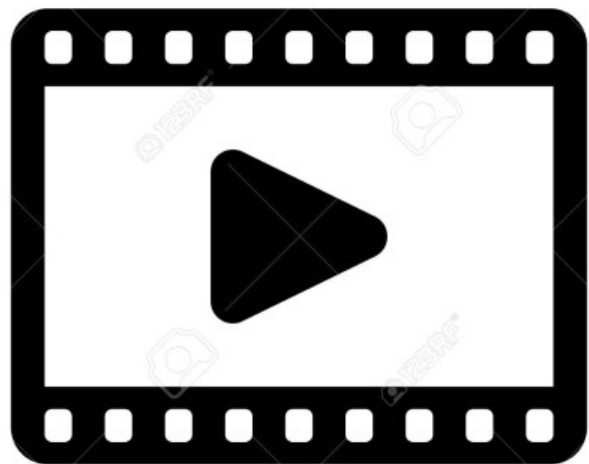
identification of **usability issues** and the
generation of **actionable suggestions**
to resolve these issues



Introduction

novel approach that uses multimodal LLMs for **automated, heuristic-based usability evaluation**

Dynamic Interaction



Video.MP4

Heuristic-Guided Evaluation

10 Usability Heuristics

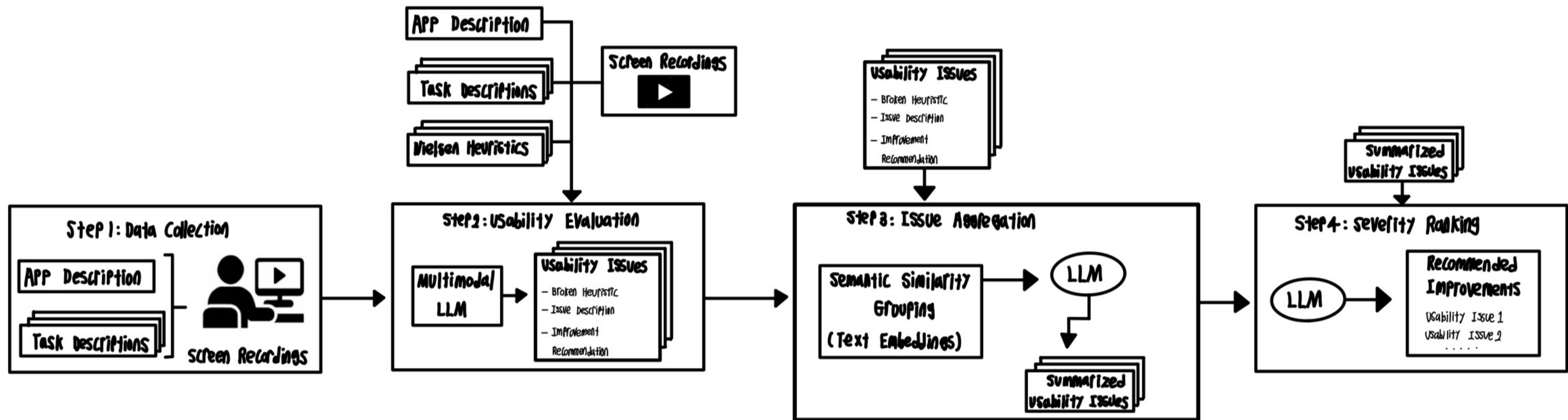
- | | |
|--|---|
|  Visibility
Show system status, tell what's happening |  Mapping
Use familiar metaphors & language |
|  Freedom
Provide good defaults & undo |  Consistency
Use same interface and language throughout |
|  Error Prevention
Helps users avoid making mistakes |  Recognition
Make information easy to discover |
|  Flexibility
Make advanced tasks fluid and efficient |  Minimalism
Provide only necessary information in an elegant way |
|  Error Recovery
Helps Users recognize, diagnose and recover from errors |  Help
Use proactive and in-place hints to guide users |

Actionable Recommendations

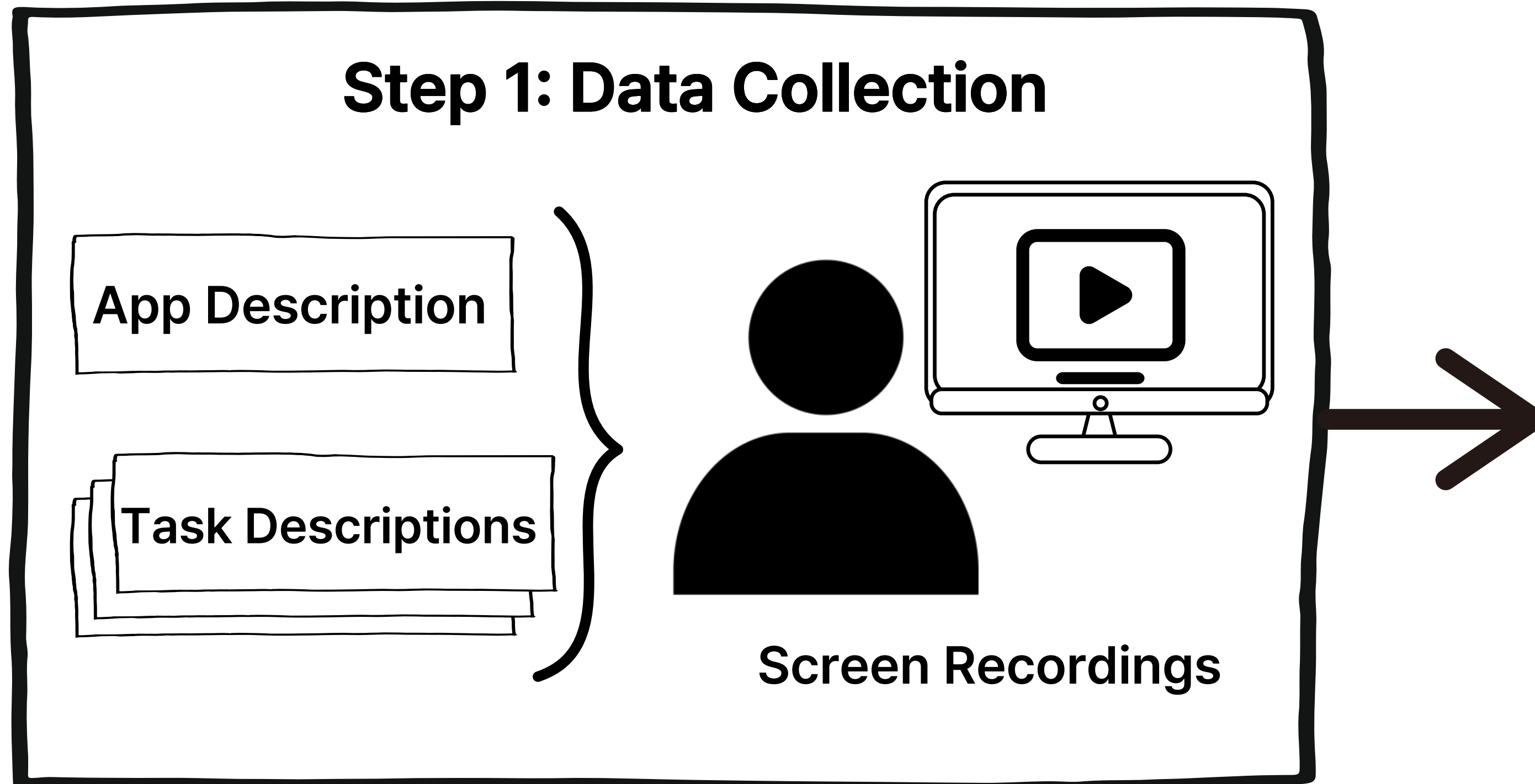
"Mask the password field with dots/asterisks (•••••), and add a toggleable 'show password' icon (👁️) so users can optionally reveal it when needed."

Approach

The study proceeds through **four phases**.

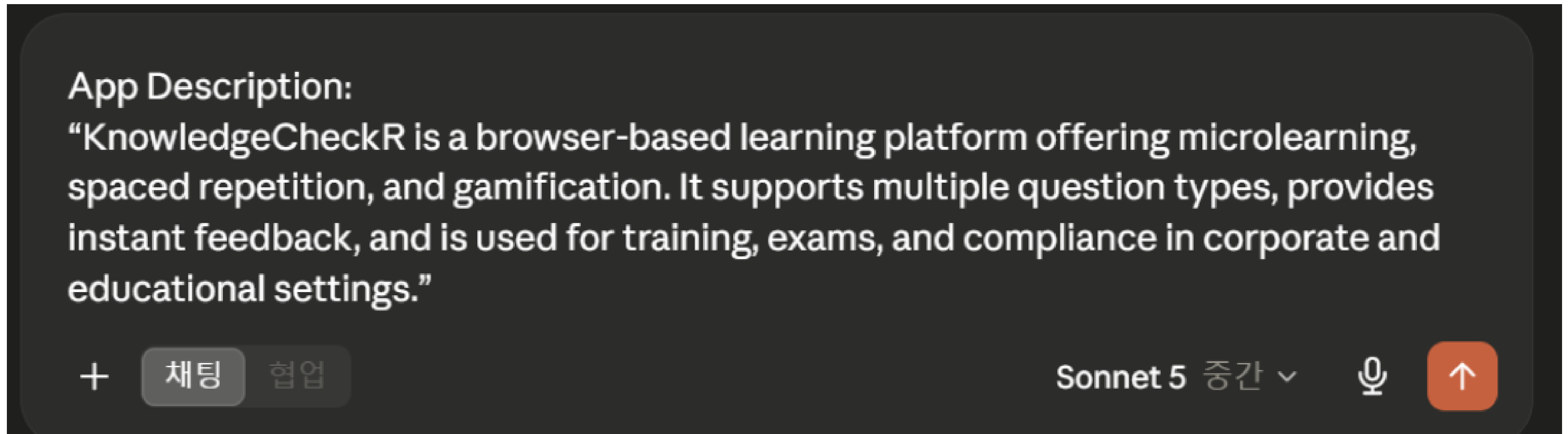


Step 1: Data Collection



Step1: Data Collection

App Description



—→ gives a general idea of the **app's purpose** to set the context

Step1: Data Collection

Task Descriptions

Assumed Persona:

“A teacher who wants to create and manage knowledge checks (quizzes) to support student learning effectively.”

+

채팅

협업

Sonnet 5 중간 ▾



Concrete task:

“Create a new knowledge check by providing a title, image, description, deadline, and adding 2 multiple-choice questions. Complete all required fields and save the knowledge check.”

+

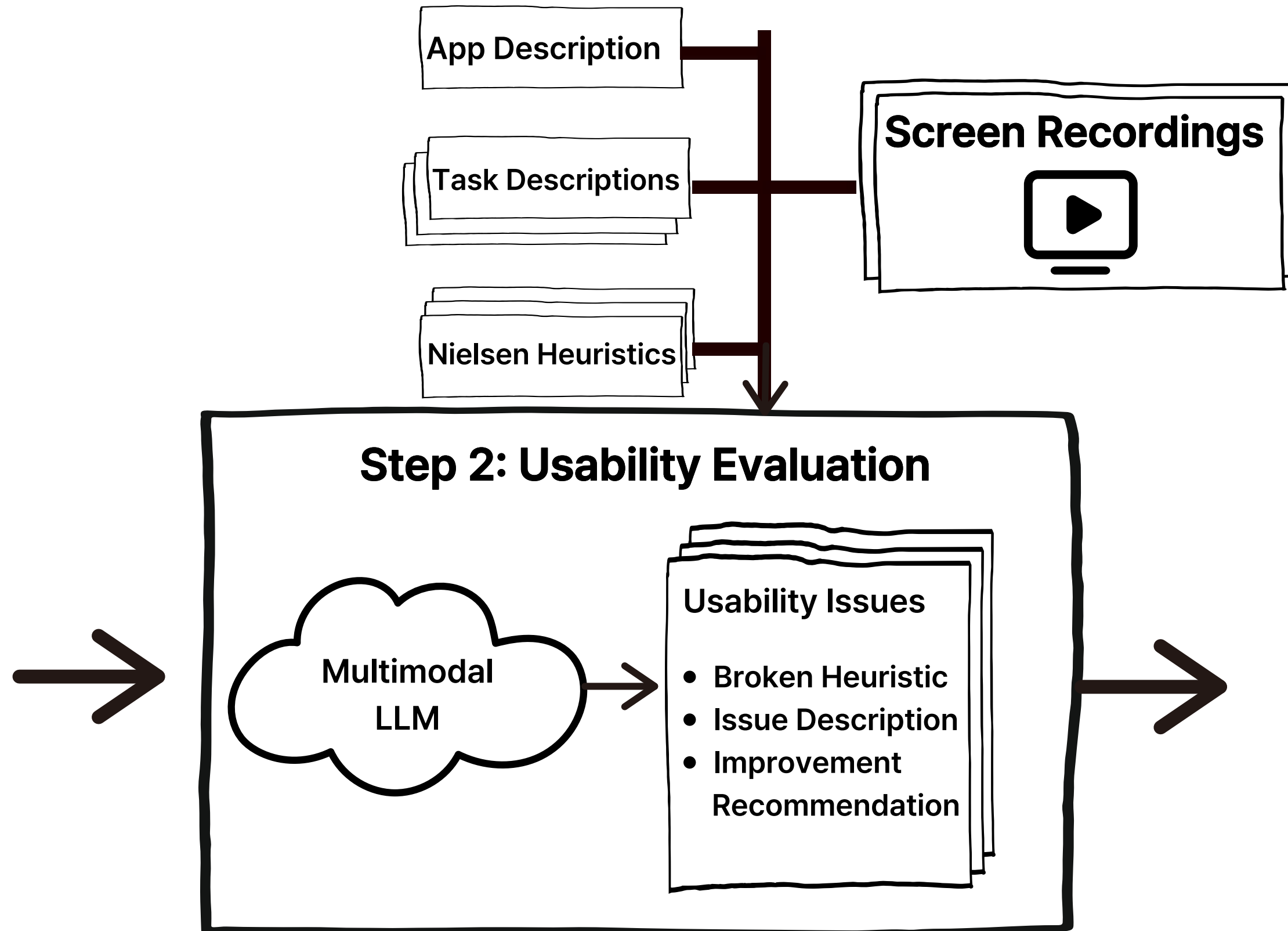
채팅

협업

Sonnet 5 중간 ▾



Step 2: Usability Evaluation



Usability Issues

- Broken Heuristic
- Issue Description
- Improvement Recommendation

Step2: Usability Evaluation

System Prompt

You are a highly experienced usability expert.
Your task is to evaluate the software interface using Nielsen's usability issues.
Use the provided user interaction recording and contextual descriptions to support a detailed, heuristic-base evaluation.
Assess each heuristic independently.
Base your evaluations strictly on the provided materials and context --- do not make assumptions.
Provided structured findings with clearly described issues and actionable recommendations.

+

채팅

협업

Sonnet 5 중간 ▾



Step2: Usability Evaluation

Prompt Template

App Description : <app_description>

User Task: <user_task>

Heuristic Under Evaluation : <heuristic_name>

Heuristic Explanation: <heuristic_explanation>

Evaluation Instructions: Evaluate the interface for the heuristic listed above. Follow these steps:

1. Identify any usability issues related to the heuristic and explain each issue.
2. Provide a practical, actionable recommendations to fix each issue.

Response Guideline:

Keep each issue's analysis and recommendation concise(1-5 sentences).

Avoid restating the heuristic definition.



Sonnet 5 중간 ▾



Step2: Usability Evaluation

Prompt Example

while analyzing the **User Control** and **Freedom heuristic**, the LLM instruction includes the description to focus on the question

Usability Heuristic(User Control and Freedom):
“Can users easily undo or exit unwanted actions (e.g., cancel, back, undo)?”



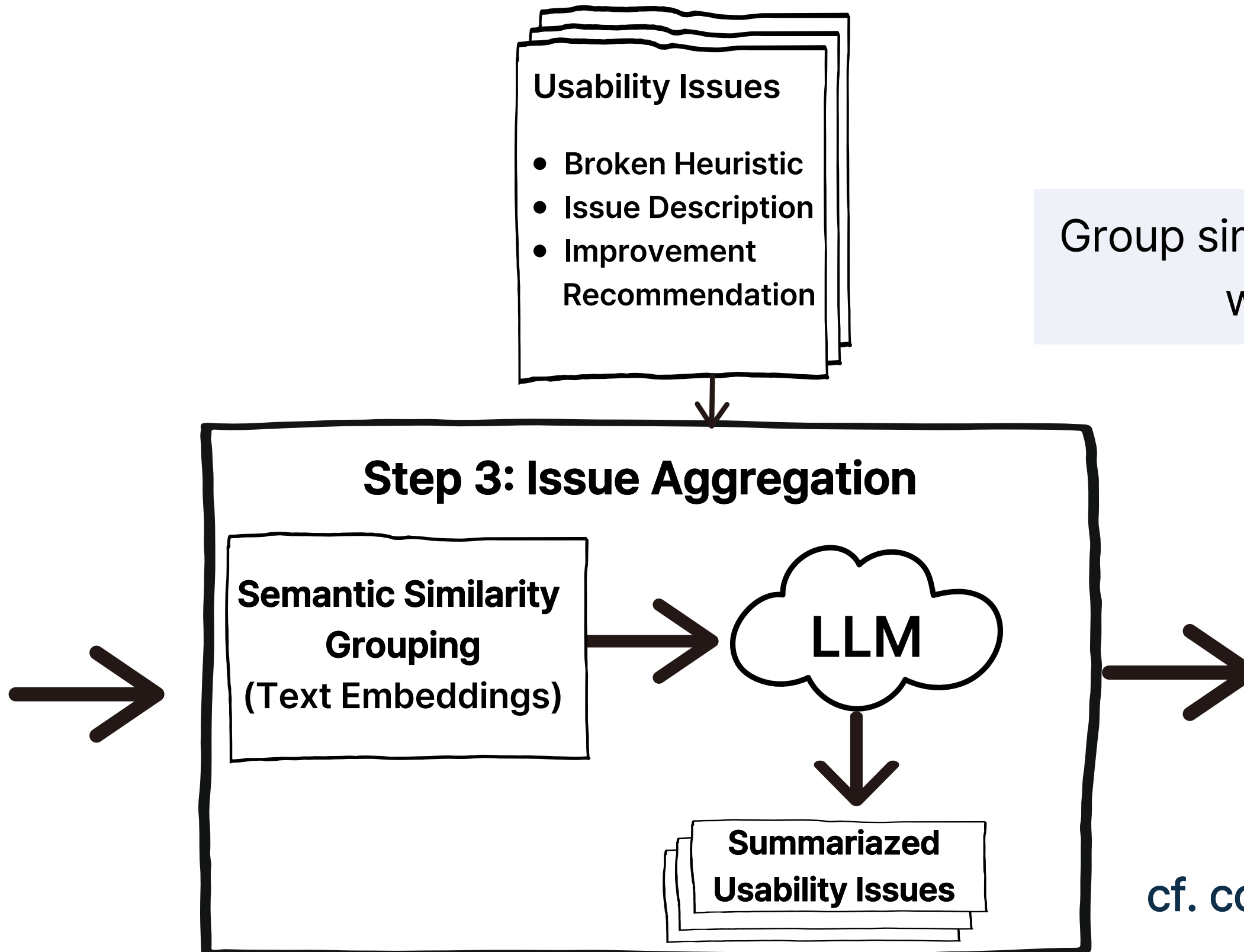
채팅

협업

Sonnet 5 중간 ▾



Step 3: Issue Aggregation



Semantic Similarity

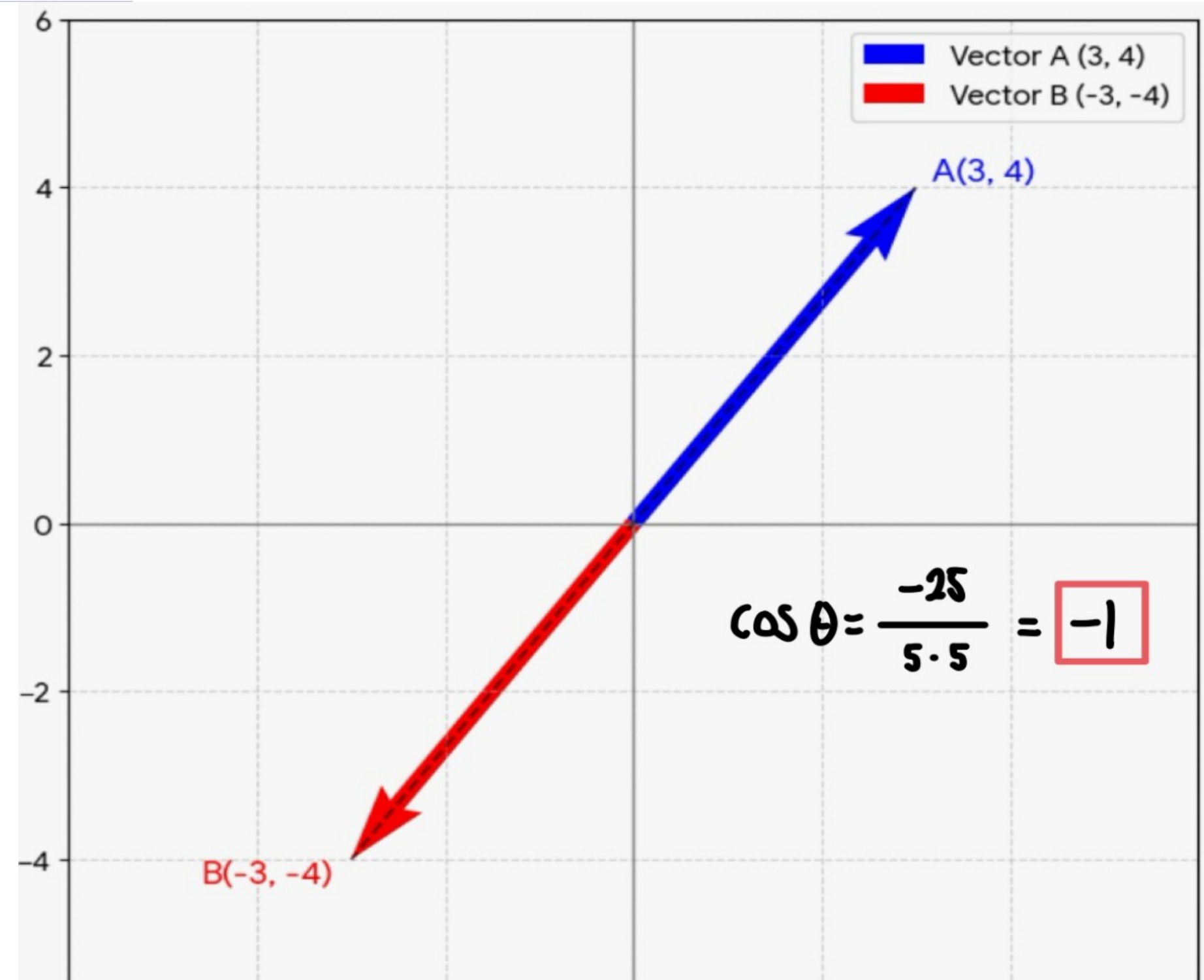
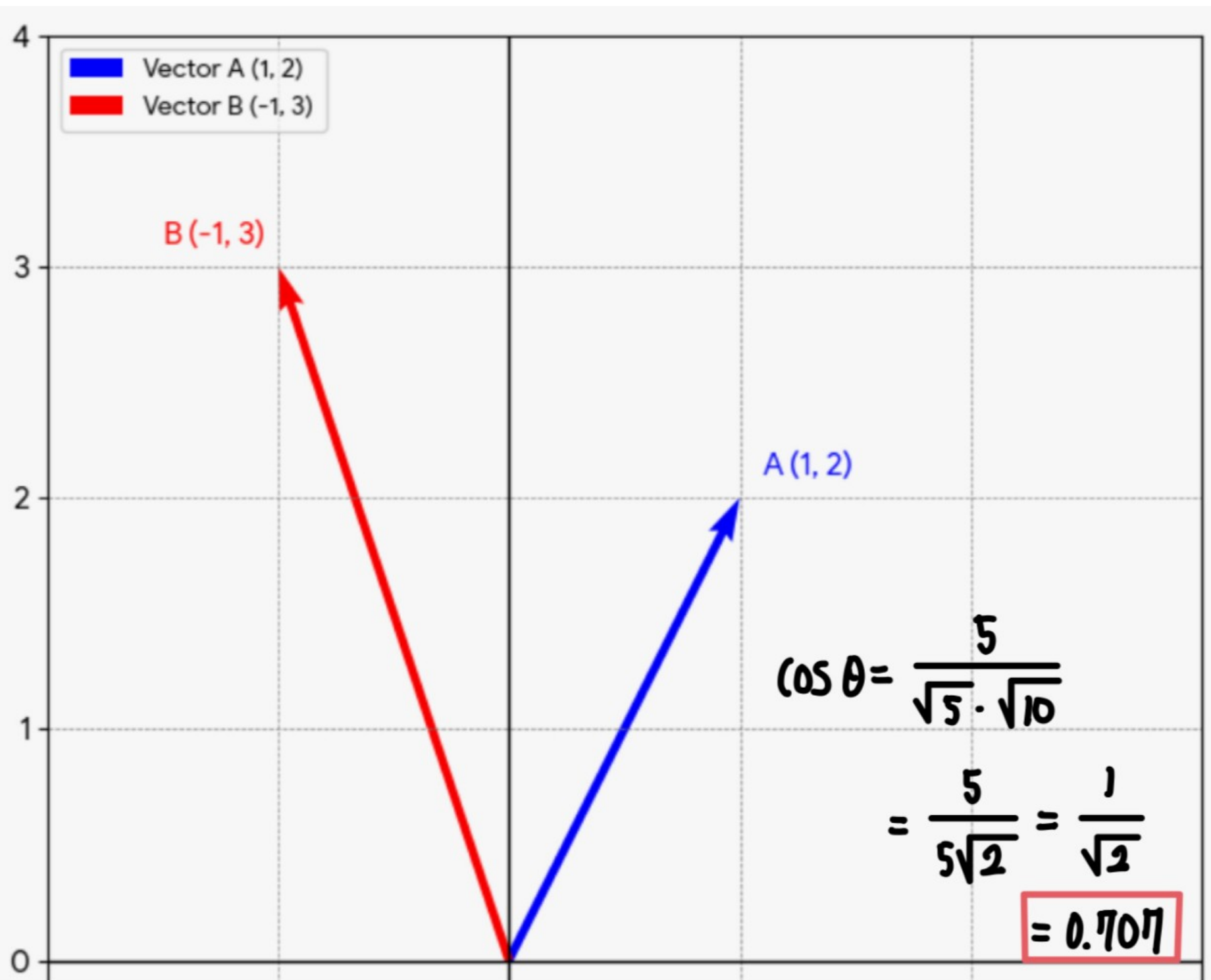
Group similar **issue descriptions** and **recommendations**, which are then summarized by an LLM

cf. cosine similarity Threshold: 0.7 !

Cosine Similarity

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \|\vec{b}\|}$$

cf. cosine similarity Threshold: 0.7 !



Step3: Issue Aggregation

Prompt template for **usability issue** summary

Summarize the following usability issue descriptions into one clear, concise problem statement(1-5 sentences) that captures the main user pain point.

Avoid repeating examples.

Focus on the core usability problem.

Descriptions:

<issue_1_description>

<issue_2_description>

<....>



Sonnet 5 중간 ▾



Step3: Issue Aggregation

Prompt template for **usability improvement recommendation** summary

Summarize the following usability improvement recommendations into one actionable suggestions that resolves the identified issues.
Combine overlapping suggestions and avoid redundancy.

Improvement Recommendations:

<issue_1_recommendation>

<issue_2_recommendation>

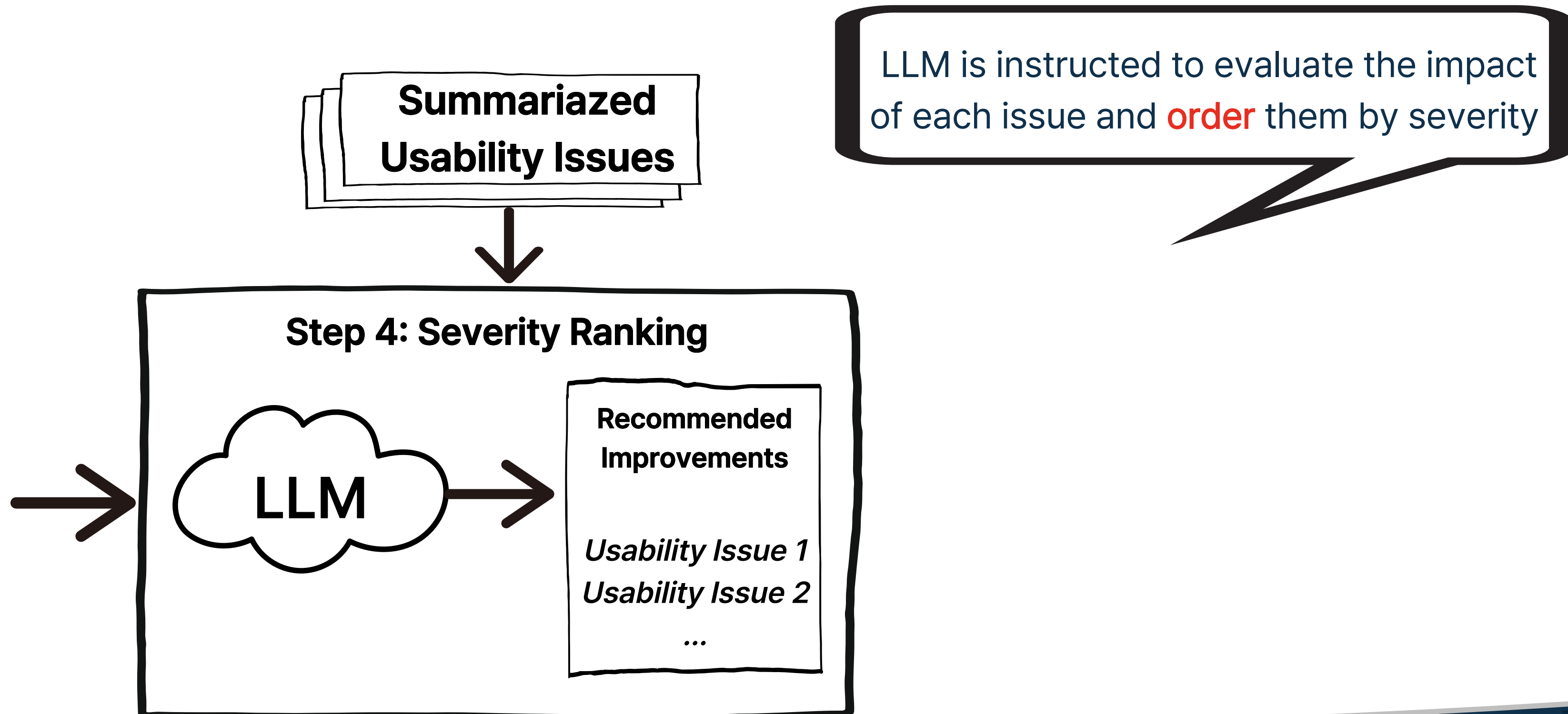
<....>

+

Sonnet 5 중간 ▾



Step 4: Severity Ranking



Step4: Severity Ranking

System prompt for the severity **ranking** of issues

You are a highly experienced usability expert applying Nielsen's heuristics.
Your task is to re-rank identified usability issues by actual severity.
Consider the impact on task success and user frustration, frequency and likelihood of occurrence, and effort required to diagnose and recover.
Base your evaluations strictly on the provided materials and context -- do not make assumptions.
Provide a ranked list from most to least severe.



Sonnet 5 중간 ▾



Step4: Severity Ranking

Prompt template for **usability issue ranking** by severity

Evaluate and re-rank the following issues by severity.
Each issue includes a description and a recommendation on how to fix the issue.

Usability issues to rank:

<usability_issue_1>

<usability_issue_2>

<....>

+

Sonnet 5 중간 ▾



Approach

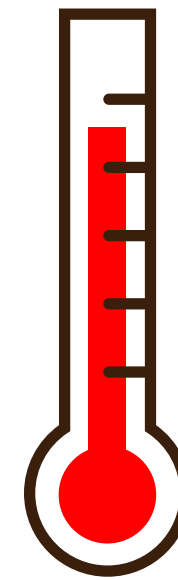
Model Details



gemini-2.0-flash-001

→ designed for efficient processing of **textual and visual inputs**

LLM Temperature : 0.0



2 -More Randomness

0 -More Deterministic

→ encourage deterministic outputs, which are suitable for **structured evaluation tasks**

1FPS (Frame Per Second)

→ each frame contributing **258 tokens**

Study Design

Application Context

1. EventHelpr



collaborative planning tool

2. KnowledgeCheckR



Learning Platform

EventHelpr			KnowledgeCheckR		
Persona	User Task	Duration	Persona	User Task	Duration
Organizer	Registration and Login	00:55	Teacher	Registration and Login	01:33
	Event Creation	01:17		Knowledge Check Creation	03:50
	Enable Decision Support	01:17		Knowledge Check Extension	01:24
	Add Initial Options	03:32		Enable Anonymized Access	00:30
Participant	Access Event	00:21	Student	Access Knowledge Check	00:18
	Contribute to Discussion	02:52		Complete Knowledge Check	01:02

Study Design

95 participants with **academic or professional backgrounds** in software engineering

- advanced master's students
- university researchers
- industry practitioners

self-reported their prior experience with usability evaluation and heuristic methods

Dimension	All (N = 95)	EHR (N = 48)	KCR (N = 47)
Usability Evaluation Experience	2.77 (1.03)	2.92 (1.06)	2.62 (0.98)
Familiarity with Usability Heuristics	2.36 (1.20)	2.40 (1.20)	2.32 (1.20)
Familiarity with Application	2.73 (1.39)	1.79 (1.21)	3.68 (0.77)

cf. EHR: EventHelpR

KCR: KnowledgeCheckR

Study Design

Methodology

First, show (a description of the evaluated application, the assigned persona, and the specific task)

- ⇒ Watch a screen recording of the task execution
- ⇒ Note any usability issues they observed
- ⇒ Evaluate **three automatically generated recommendations** that were considered **most severe** for the given scenario

Generated Recommendations

- Description of the identified usability issue
- Actionable improvement suggestion

Study Design

Methodology

Item-level evaluation metrics used to assess **each usability recommendation**

Metric	Description	Response Type	RQ
Understandability	Clarity of the issue and improvement suggestion	5-point Likert scale	RQ1
Plausibility	Relevance of the issue and suggestion to the task	5-point Likert scale	RQ1
Completeness	Coverage of relevant usability issues	3-point scale & open text	RQ1
Awareness	Whether participant would have detected the issue independently	Binary (Yes/No)	RQ2

Study Design

Methodology

After completing the review of all three usability improvement recommendations, the participants provided **general feedback**

General feedback questions asked after the recommendation reviews

Metric	Description	Response Type	RQ
Helpfulness	Helpfulness of recommendations for identifying	5-point Likert scale	RQ2
Perceived Complexity	Difficulty of evaluating usability without assistance	5-point Likert scale	RQ3
Practical Usefulness	Usefulness of the system in real-world UI/UX	5-point Likert scale	RQ3
Open Feedback	Additional participant comments	Open text	Exploratory

Results

RQ1. Quality of Usability Improvement Recommendations

- (1) How **clearly** are the recommendations articulated
- (2) How **plausible** are the connections between the presented issues and the given screen recordings and tasks

Dimension	Mean	Median	Standard Deviation
Issue Clarity	4.21	4	0.91
Recommendation Clarity	4.28	4	0.92
Issue Plausibility	4.04	4	1.00
Recommendation Plausibility	4.10	4	1.00

Results

RQ1. Quality of Usability Improvement Recommendations

(3) How **complete** are the recommendations in covering relevant usability issues

Completeness Rating	Count	Percentage
Complete	210	73.7%
Partially Complete	65	22.8%
Incomplete	10	3.5%

Results

Spearman correlation between **participant's ratings of the recommendations** (clarity, plausibility, and completeness) and 1) **experience** in usability evaluation, and 2) **familiarity** with usability heuristic

Evaluation Metric	Experience	Heuristic Familiarity
Issue Clarity	-0.0131	-0.0730
Recommendation Clarity	0.0256	-0.0104
Issue Plausibility	0.0098	-0.0694
Recommendation Plausibility	-0.0438	-0.0884
Recommendation Completeness	-0.0186	-0.0721

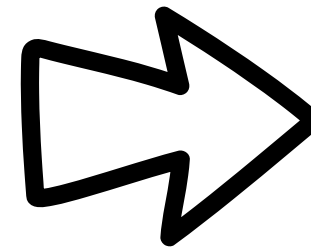
➡ Generated recommendations were understandable and actionable **regardless of expertise level**

Results

RQ2. Perceived Helpfulness for Usability Issues Detection

Participants ratings of how helpful the recommendations were to identify usability issues

Rating	Count	Percentage
1	1	1.1%
2	5	5.3%
3	19	20.0%
4	47	49.5%
5	23	24.2%
Mean	3.91	
Median	4	
SD	0.86	



The automatically generated feedback supported users in recognizing relevant usability issues, even **without requiring deep domain knowledge**

Results

RQ2. Perceived Helpfulness for Usability Issues Detection

Responses to whether participants would have identified the usability issues without system prompt

Response	Count	Percentage
Yes	137	48.1%
No	94	33.0%
I don't know	54	18.9%

Results

RQ3. Perceived Complexity of Usability Evaluation

Participant ratings on perceived evaluation complexity without assistance (1: Very Easy / 5: Very Difficult)

System usefulness in UI/UX workflows (1: Not at all / 5: Extremely)

Metric	Perceived Complexity	System Usefulness
1	13 (13.7%)	3 (3.2%)
2	35 (36.8%)	4 (4.2%)
3	30 (31.6%)	17 (17.9%)
4	15 (15.8%)	43 (45.3%)
5	2 (2.1%)	28 (29.5%)
Mean	2.56	3.94
Median	2	4
SD	0.98	0.96

≥ 50%

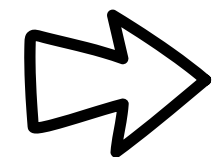
) 75%

Results

RQ3. Perceived Complexity of Usability Evaluation

Spearman correlations between participants' self-reported usability evaluation experience and familiar with usability heuristics, and their ratings of perceived evaluation complexity and system usefulness

Evaluation Metric	Experience	Heuristic Familiarity
Perceived Complexity	0.1201	0.1166
System Usefulness	0.0224	0.0612

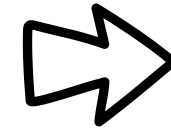


Possibly due to their **greater awareness** of what a thorough evaluation entails

Discussion

Challenge1: Balance detail and simplicity

Flexible Presentation formats



usability Issue Summary

Search button is not prominent.
Users may overlook the search function.

On demand



Challenge2: Extension of the approach with domain-specific heuristic



ex) Banking App

Specific Heuristic: Transaction Confirmation

Challenge3: Aggregate results from multiple MLLM



+



+



Gemini

+

o1

Challenge4: Release an accessible tool



Plugins

Discussion

Limitation

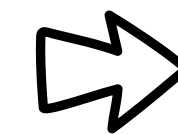
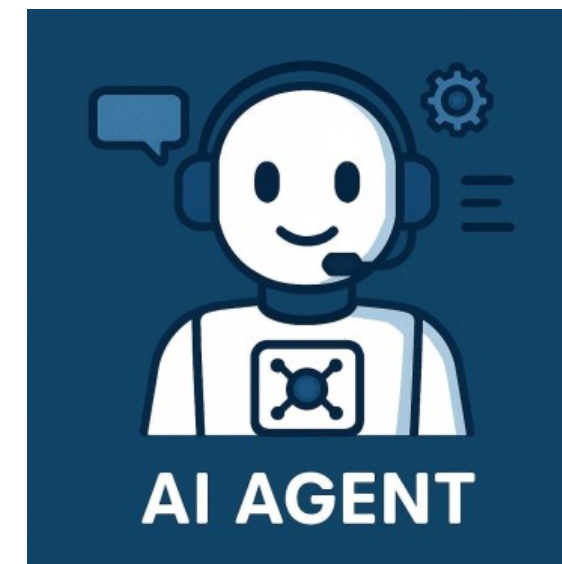
Manual definition of user tasks and the recording of interaction videos

Future Work

Extracting user tasks from project artifacts



Using intelligent agent
to **simulate** user behavior and generate recordings



「● REC」

Threats to Validity

1. Evaluation was conducted with **two applications**



2. Ratings of clarity, plausibility, and completeness are **subjective**

3. Results depend on **specific modeling choices**, including the selected embedding model, MLLM, and prompt design



Thank you!

**Recommending
Usability Improvements with Multimodal
Large Language Models**

Base Lab @DGIST